

Part A. 1) $x(t) = m(t) + c(t)$

$$\begin{aligned}
 y(t) &= a x(t) + b x^2(t) \\
 &= a (m(t) + c(t)) + b (m^2(t) + c^2(t) + 2m(t)c(t)) \\
 &= \underbrace{a c(t) + 2b m(t)c(t)}_{\text{Conventional AM}} + \underbrace{a m(t) + b m^2(t) + b c^2(t)}_{\text{additional terms}}
 \end{aligned}$$

Additional terms need to be eliminated. Hence use bandpass filter centered at f_c with bandwidth B .
 $\ast a m(t)$ filtered out as $f_c \gg B$.

$\ast b m^2(t)$ " " " $f_c \gg 2B$

$\ast b c^2(t)$ " " " $2f_c$ much higher than B .

Conventional AM signal: $a c(t) + 2b m(t) c(t)$

$$= a \left[1 + \frac{2b}{a} m(t) \right] c(t)$$

$$\mu = \frac{2b}{a} \cdot \max(m(t)) = \frac{2b}{a} \cdot A_m$$

$$V_{AM}(t) = A_c (1 + \mu m_n(t)) \cos 2\pi f_c t, \quad \mu = \frac{A_{\min}(m(t))}{A_c}$$